
Lab 2: Parity bit

Goals

- The objective of this lab is to design basic combinational circuits using logic gates.
- Familiarize yourself with the Quartus Prime Lite development environment.

Introduction

In this lab, a mechanism will be implemented to determine the **parity bit**.

The **parity bit** serves as an error detection mechanism in digital systems. Its primary purpose is to verify data integrity during transmission or storage.

When data is transmitted or stored along with its parity bit, the receiver can recalculate the parity of the data and compare it with the received parity bit. If there is a discrepancy, an error in transmission or storage is detected.

Lab description

There are two types of parity:

- **Even Parity:** The total number of '1' bits in the word must be even.
- **Odd Parity:** The total number of '1' bits in the word must be odd.

Considering an **8-bit word**, 7 bits represent data, and the 8th bit is the **parity bit**. The following table presents examples of words including the parity bit:

7 bit data	Count of 1's	8 bit word	
		Even parity	Odd parity
0000000	0	0000000 0	0000000 1
1001001	3	1001001 1	1001001 0
1100110	4	1100110 0	1100110 1
1111111	7	1111111 1	1111111 0

The implemented circuit must meet the following specifications:

- It must have **7 data bits as input**, using the FPGA **switches** to represent the data bits.
- It must include an input to select the **parity type**:
 - **0** for even parity
 - **1** for odd parity
- The circuit output must be an **8-bit word** (the 7 data bits plus the parity bit). The **8 bits** will be displayed using LEDs.

Step 1

Design a circuit to determine the **parity** of the input data.

Step 2

Design a circuit to determine the **parity bit**, depending on the input data parity and the selected parity type.

Step 3

Implement the circuit on the **FPGA**.

- Use **switches SW[1] to SW[7]** for data input.
- Use **switch SW[17]** to select the parity type.
- Use **LED LEDR[0]** for the parity bit and **LEDs LEDR[1] to LEDR[7]** to display the input data.

Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

Video:

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.